

Recursive Task Composition

1 Seed tasks

Atomic tasks from existing data sources.

T1 East sales



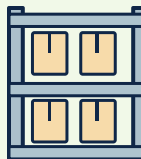
SKU	Units
SKU A	30

T2 East stock



SKU	Units
SKU A	12

T3 West stock



SKU	Units
SKU A	25

Same SKU, same date

2 Recursive candidates

Compose new tasks from existing ones (with reuse).

Round 1: compose gap computation

T1

East sales
(30 units)

T2

East stock
(12 units)

T12: East gap

$30 - 12 = 18$

Reuse candidate

Round 2: compose transfer plan

T12

East gap
(18 units)

T3

West stock
(25 units)

T123: Transfer plan

Use $\min(18, 25) = 18$ units

3 Reconstruct and validate

Build an executable plan and verify it meets the request.



User request

Cover East's sales gap using available West stock for SKU A.

Executable reference graph (reconstructed)



Read East sales

SKU A: 30



Read East stock

SKU A: 12



Read West stock

SKU A: 25



Gap = 18

$30 - 12 = 18$



Transfer = $\min(18, 25) = 18$



Transfer 18 units ✓

Validation (acceptance checks)



Replay

Re-execute steps



Goal checks

Satisfies request



Final verification

Outputs consistent

Illustrative example; not a dataset record. Reconstruction may merge or revise steps.